Circular Statistics — Quick Reference

SEAC 2026 Workshop: Circular Statistics in Archaeoastronomy and Ethnoastronomy Using R

1 Why circular statistics?

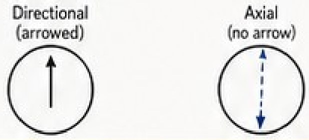
Directions repeat after one complete turn. On a 0–360 degree scale, 359 degrees and 1 degree are only 2 degrees apart, even though ordinary arithmetic treats them as far apart.

Teaching example:
358, 359, 1, 2 degrees

Ordinary arithmetic mean = 180 degrees (South).
Circular mean ≈ 0/360 degrees (North).

2 Directional or axial?

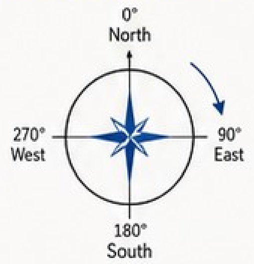
- **Directional data:** start and end matter. Example: orientation measured from a church entrance toward the apse.
- **Axial data:** the axis has no arrow. An axis at 20 degrees is equivalent to the same axis at 200 degrees.



⚠ Never analyse axial measurements as ordinary 0–360 degree directional data without an appropriate transformation.

3 Geographic azimuth convention

- 0 degrees = North
- 90 degrees = East
- 180 degrees = South
- 270 degrees = West
- *Angles increase clockwise*



Always document the convention used.

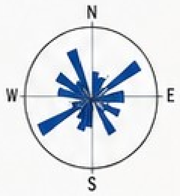
4 Circular object in R

```r
library(circular)

az <- circular(
   dat$azimuth_deg,
   units = "degrees",
   template = "geographics",
   modulo = "2pi"
 )
```

5 Rose diagram

```r
rose.diag(
  az,
  bins = 18,
  radii.scale = "sqrt"
)
```



Plot first, test second. Try more than one defensible bin count.

6 Mean direction

```r
mean_deg <- as.numeric(mean(az)) %% 360
```

Use only when a single central direction is meaningful. A pooled or multimodal sample can make one mean misleading.

7 Mean resultant length (R-bar)

```r
Rbar <- rho.circular(az)
```

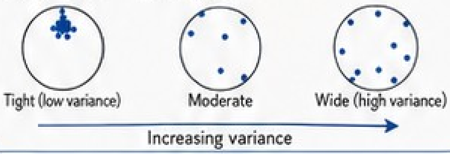
- **Near 1:** *strong net directional concentration*
- **Near 0:** *weak net directional concentration*

ℹ *R-bar near 0 can result from uniformity or from opposing clusters that cancel vectorially.*

8 Circular variance

```r
circular_variance <- 1 - Rbar
```

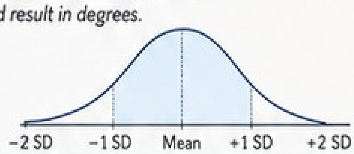
Near 0 = tighter concentration; larger values = more dispersion.



9 Circular standard deviation

```r
circular_sd_deg <- sd(az) * 180 / pi
```

The explicit conversion keeps the reported result in degrees.



10 Confidence interval for mean direction

```r
set.seed(2026)

mean_ci <- mle.vonmises.bootstrap.ci(
  az,
  alpha = 0.05,
  reps = 2000
)
```

Use the interval as a measure of precision for the estimated mean direction under the bootstrap/model assumptions. It does not measure archaeological certainty.

11 Rayleigh test

```r
rayleigh_result <- rayleigh.test(az)
Rbar <- rho.circular(az)
z <- length(az) * Rbar^2
p <- rayleigh_result$p.value
```

H0: *directions are uniform around the circle*
General alternative: *unimodal directional concentration*

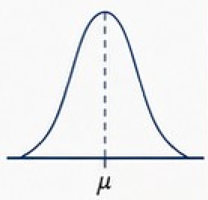
ℹ *A small p-value is evidence against circular uniformity for that alternative. A non-significant result does not prove uniformity, especially for bimodal or multimodal patterns.*

12 von Mises model

The von Mises distribution is a useful single-cluster circular model.

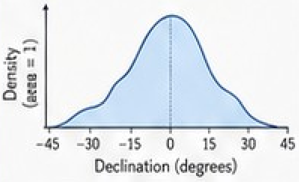
- **mu** = central direction
- **kappa** = concentration; larger values indicate tighter concentration

```r
vm <- mle.vonmises(az)
as.numeric(vm$mu) %% 360
vm$kappa
```



⚠ Do not force a single von Misee model onto an obviously multimodal sample.

13 Curvigram baseline

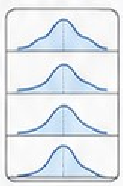


Coordinate:	declination_deg
Geometry:	linear
Kernel:	Epanechnikov
Bandwidth:	2 × uncertainty_deg
Scaling:	area = 1
Range:	−45 to +45 degrees
Step:	0.05 degrees

ℹ uncertainty_deg *is the reported ± measurement uncertainty. It is not automatically assumed to be one standard deviation.*

14 Curvigram sensitivity checks

- ✓ Epanechnikov × 1
- ✓ Epanechnikov × 2 (baseline)
- ✓ Epanechnikov × 3
- ✓ Gaussian × 2



Ask whether the substantive feature stays in approximately the same place and remains visible.

15 Three rules to remember

1. **Plot first, test second.**
2. **Match the method to the distribution and research question.**
3. **Statistical concentration is not proof of cultural or astronomical intention.**